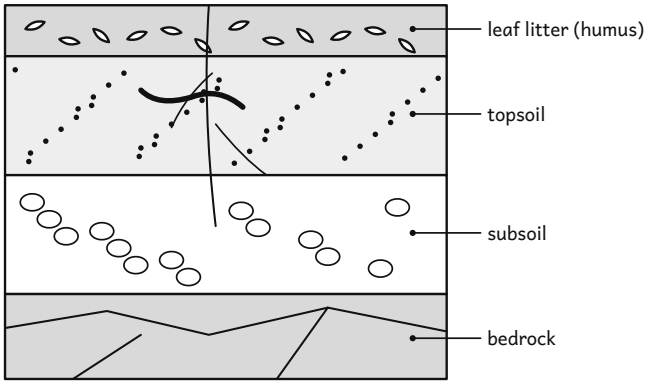
EARTH AND SPACE

Soil and the Living Earth

What soil is, soil types and a key to identify them, earthworms and plant growth



In this pack

Learn it: Soil: the living skin of the Earth ★☆☆
Fact check: Soil layers and soil types ★☆☆
Think it through: Soil, worms and plants ★★☆
Investigate: Which soil holds the most water? ★★☆
Real-world case study: Worms that make money: vermicompost ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · B2 · SOIL

Soil: the living skin of the Earth

★☆☆
Grade 5

Name: Class: Date:

Learning goals

- Describe what soil is made of and how it forms.
- Use a simple key to identify soil types.
- Explain what earthworms and plants need from soil.

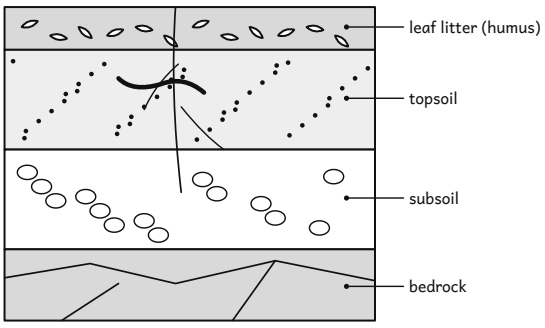
Almost all our food grows in a thin layer of **soil**. In many places this layer is only a few tens of centimetres deep, yet it can take hundreds of years to form just a few centimetres of new soil. Soil is not just "dirt": it is a mixture full of life.

What is soil made of?

- tiny pieces of **rock** (sand, silt and clay)
- **humus**: dark, rotted remains of dead plants and animals
- **water** and **air** in the spaces between the particles
- **living things**: earthworms, insects, fungi and millions of bacteria

How does soil form?

Rain, ice, heat and plant roots slowly break rocks into small pieces. This is called **weathering**. Dead leaves and animals are broken down by **decomposers** into humus, which mixes with the rock pieces. Rivers also carry soil and drop it in new places.



If you dig a deep hole you can see layers.

Types of soil

Soil type	What it is like
sandy	large, gritty particles; water drains through fast; dries out quickly; few nutrients
clay	tiny particles; sticky when wet, cracks when dry; holds water; can be rich in nutrients
chalky	pale and stony; contains chalk, so it fizzes with vinegar; drains well
woodland	dark and crumbly; full of rotting leaves (humus); holds water well
garden topsoil	a mixture of sand, clay and humus; dark and crumbly; good for most plants

Bangladesh connection

Most of Bangladesh is a giant river **delta**. Every year the Ganges, Brahmaputra and Meghna rivers drop fine soil called **silt** on the land. This makes the soil very fertile, which is why farmers can grow two or three rice crops a year.

Earthworms: soil engineers

Earthworms need **moist, loosely packed** soil. They have no lungs and breathe through their damp skin, so dry soil kills them. As they tunnel they let air and water into the soil. They eat dead leaves and soil, and their droppings (**worm casts**) are full of nutrients. Charles Darwin studied earthworms for 40 years!



An earthworm. The thick band helps it make eggs.

Plants and soil

Plants grow best in soil that is **moist** and has plenty of **nutrients** (minerals). Roots also need air. Very sandy soil dries out and loses its nutrients; heavy clay can become waterlogged. Adding **compost** improves both kinds.

Key words

soil	mixture of rock particles, humus, water, air and living things
humus	rotted remains of plants and animals in soil
weathering	rocks breaking into smaller pieces
silt	very fine soil carried by rivers
drainage	how fast water flows through soil
nutrients	minerals in soil that plants need
compost	rotted plant waste added to soil to improve it
key	a set of questions used to identify things

FACT CHECK · B2 · SOIL

Soil layers and soil types

★☆☆
Grade 5

Name: Class: Date:

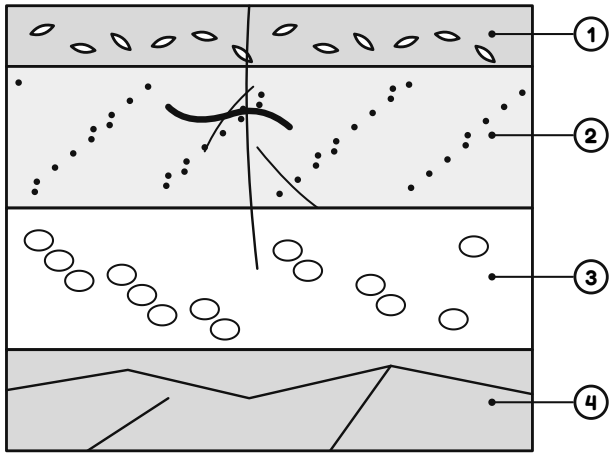
Learning goals

- Label soil layers.
- Use a key to identify soils.

1 **Label the layers of soil.** [4]

LABELS

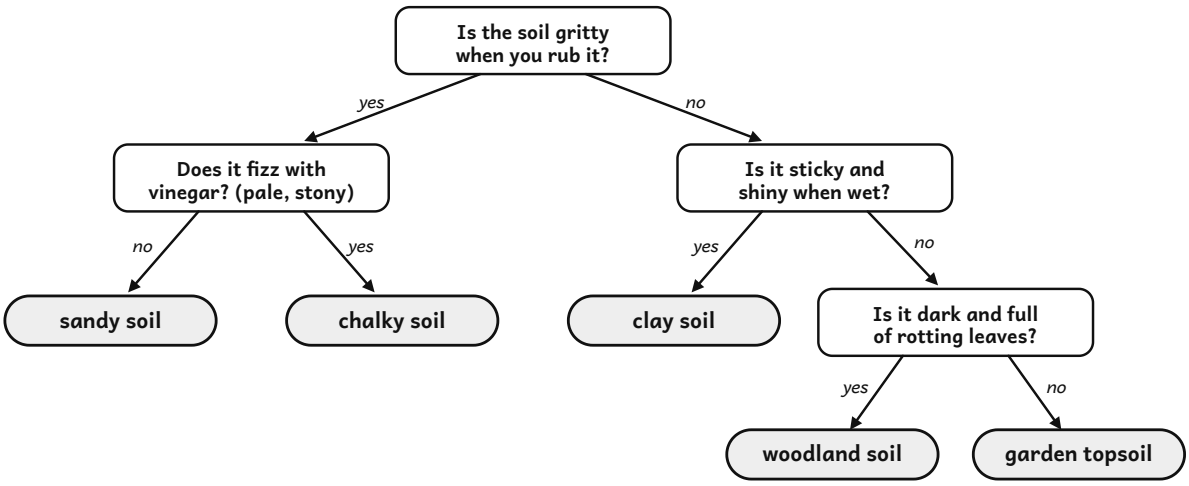
subsoil **leaf litter (humus)** **bedrock** **topsoil**



1 _____ 2 _____

3 _____ 4 _____

2 **Use the key to identify each soil sample.** [5]



Sample	What the class observed	Soil type
A	gritty, does not fizz	
B	not gritty, sticky and shiny when wet	
C	not gritty, not sticky, dark with rotting leaves	
D	gritty, pale and stony, fizzes	
E	not gritty, not sticky, dark and crumbly, few leaves	

3 **Fill in the gaps.** [5]

WORD BANK

plastic **air** **dry** **nutrients** **skin** **humus** **moist**

Soil is a mixture of rock particles, _____, water and _____. Earthworms need _____ soil because they breathe through their _____. Plants grow best in moist soil with plenty of _____.

THINK IT THROUGH · B2 · SOIL

Soil, worms and plants

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Explain soil properties.
- Link soil to plant growth and soil animals.

1 **Why does water drain quickly through sandy soil but slowly through clay?** [2]

2 **Rina wants to grow vegetables, but her soil is very sandy. Suggest what she could do and explain why.** [2]

3 **Earthworms come up to the surface after heavy rain. Suggest why.** [2]

4 **Explain two ways earthworms help plants.** [2]

5 **Why does the soil of the river delta make Bangladesh good for farming?** [2]

INVESTIGATE · B2 · SOIL

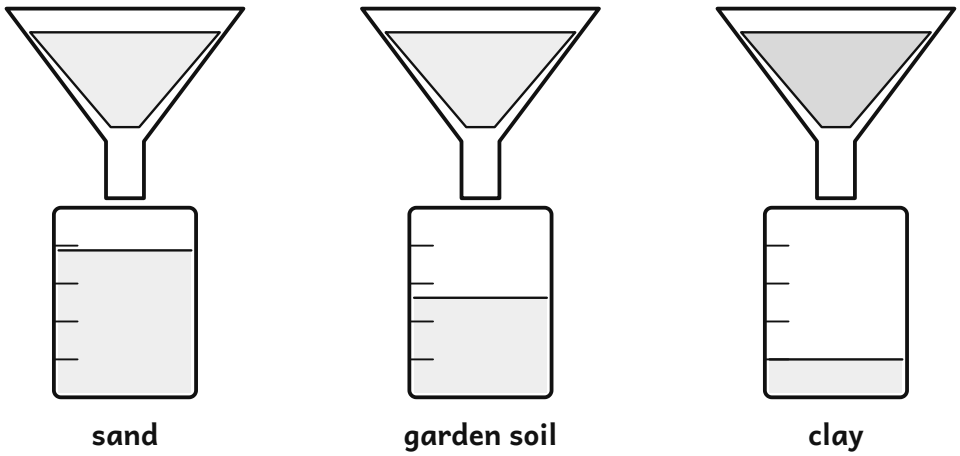
Which soil holds the most water?

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Carry out a fair test.
- Present results as a bar chart.
- Use results to answer a real question.



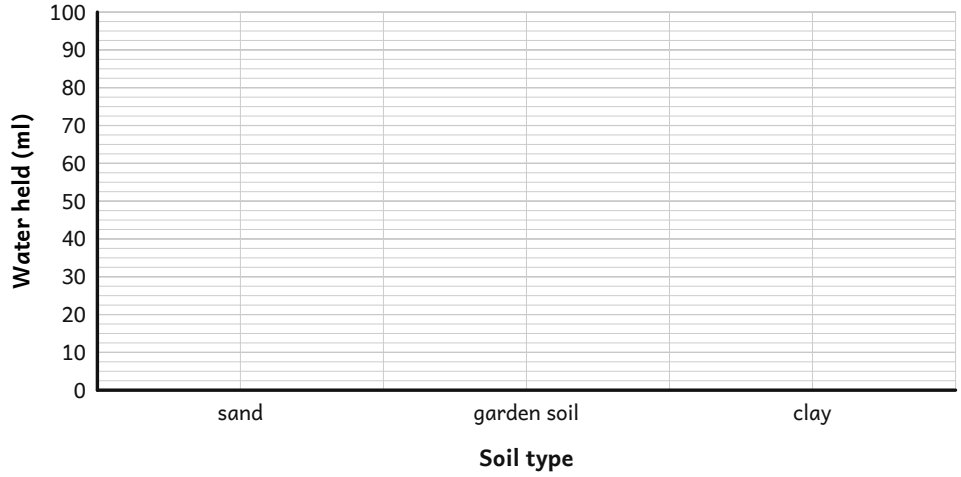
Equal amounts of soil in each funnel; water poured in at the same time.

Class 5C put 100 g of each soil in a funnel lined with filter paper. They poured 100 ml of water into each one and measured how much water dripped through in 5 minutes.

1 **Complete the table to work out how much water each soil held.** [3]

Soil	Water poured in (ml)	Water that dripped through (ml)	Water held by soil (ml)
sand	100	80	
garden soil	100	55	
clay	100	20	

2 **Draw a bar chart of the water held by each soil.** [4]



3 **Name three things the class kept the same to make it a fair test.** [3]

4 **Which soil would be best for a plant that needs a lot of water, and which would be best for a cactus? Explain.** [2]

REAL-WORLD CASE STUDY · B2 · SOIL

Worms that make money: vermicompost

★★★ Grade 5

Name: **Class:** **Date:**

Learning goals

- Apply soil science to farming.
- Use data to compare.
- Consider benefits and costs.

Science · Economics · Maths · Environment

In many villages of Bangladesh, farmers, many of them women, keep earthworms in concrete rings or large pots filled with cow dung, vegetable peel and straw. The worms eat the waste and turn it into rich, dark **vermicompost** (*kecho sar*) in about 6 to 8 weeks.

Farmers use vermicompost on their own fields and sell the rest. Many say their vegetables grow better and they can use less **chemical fertiliser**. Chemical fertiliser gives quick nutrients, but using too much can harm soil life and wash into rivers and ponds.

Based on vermicompost projects in Bangladesh; numbers below are examples for learning.

1 **Why does vermicompost help plants grow?** [2]

2 **Fatema sells vermicompost for 15 taka per kilogram. One worm ring makes 40 kg every 2 months.** [3]

a) How much money does one ring earn in 2 months?

b) She has 5 rings. How much do they earn in 2 months?

c) How much in one year (6 lots of 2 months)?

3 **A farmer tested two plots of tomatoes (example results).** [3]

Plot	Fertiliser	Tomatoes harvested (kg)
A	chemical fertiliser only	120
B	half chemical + vermicompost	135

a) Which plot grew more tomatoes and by how much?

.....

b) Is one test enough to be sure? Explain.

.....

.....

4 **Give two benefits of vermicompost for the environment.** [2]

.....

.....

REVIEW AND CHALLENGE · B2 · SOIL

Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

L	U	K	B	P	C	H	A	L	K
I	N	M	G	E	C	C	M	P	D
W	S	J	V	P	O	W	O	R	M
V	N	O	A	L	M	G	L	F	R
O	S	O	I	C	P	D	Y	G	F
D	I	U	A	L	O	M	H	H	V
C	L	G	I	T	S	S	U	O	N
L	T	K	P	C	T	A	M	I	H
A	F	D	F	M	O	N	U	E	K
Y	R	O	C	K	E	D	S	T	I

chalk clay compost humus rock sand silt soil worm

2 **Which soil is sticky when wet and cracks when dry?** [1]

- ◯ sandy
- ◯ clay
- ◯ chalky
- ◯ woodland

3 **Which soil would fizz with vinegar?** [1]

- ◯ sandy
- ◯ clay
- ◯ chalky
- ◯ garden topsoil

4 **Design a "worm hotel" to keep earthworms healthy for one week in the classroom. Draw and label it.** [3]

My worm hotel

5 **Challenge. Soil takes hundreds of years to form but can be washed away in one flood. Why should we protect it?** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
describe what soil is made of	○	○	○
use a key to identify five soil types	○	○	○
explain why earthworms need moist, loose soil	○	○	○
explain why plants grow better in moist, nutrient-rich soil	○	○	○
carry out a fair test on soil drainage	○	○	○
explain the benefits of compost	○	○	○